

Morphic Studio

Document 20 — Comic Studio (Expanded Specification)

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Version: 2.0

Status: Expanded Specification

Priority: Highest (Core Production Module)

Purpose

The Comic Studio is the primary production environment for creating, editing, and publishing AI-assisted comics within Morphe Studio.

It transforms approved storyboards into fully editable comic pages while preserving narrative consistency, visual continuity, and creator control.

Rather than replacing creators, the Comic Studio acts as an intelligent production partner, automating repetitive tasks while allowing creators to direct every artistic decision.

Vision

The Comic Studio should enable one creator to produce professional-quality comics that would traditionally require an entire creative team.

The creator remains the director.

The AI becomes the production team.

Every generated panel should be editable, reusable, and connected to the project's Story Bible, Character Manager, World Builder, Asset Library, and Creative Memory Engine.

Core Philosophy

Generate Once. Edit Forever.

Generated content is only the beginning.

Creators should be able to refine, replace, rearrange, and improve every element without rebuilding the entire page.

Objectives

The Comic Studio shall:

- Generate comic pages from approved storyboards.
 - Maintain character consistency.
 - Maintain environmental consistency.
 - Reuse existing assets whenever possible.
 - Provide professional editing tools.
 - Support multiple comic layouts.
 - Support multiple publishing formats.
 - Synchronize with every core Morpheic Studio system.
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Production Pipeline

The Comic Studio receives:

Approved Story

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Approved Storyboard

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Character Data

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World Data

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Asset Library



Creative Memory



AI Planner



Comic Production



Creator Review



Final Approval



Publishing

No comic generation begins without an approved storyboard unless the creator explicitly chooses to bypass it.

Comic Structure

Projects are organized as:

Project



Volume (Optional)



Chapter

↓

Page

↓

Panel

↓

Dialogue

↓

Assets

↓

Metadata

Every panel is independently editable.

Panel Generation

For every panel, the system shall determine:

- Characters
- Outfits
- Facial expressions
- Body poses
- Camera angle
- Camera distance
- Lighting
- Weather
- Background
- Props
- Dialogue
- Sound effects
- Visual effects

AI must reference approved project data before generating new content.

Comic Layout System

Supported layouts include:

- Traditional Western Comics
- Manga
- Webtoon (Vertical Scroll)
- Graphic Novel
- Newspaper Strip
- Four-Panel Comic
- Custom Layout

Creators may define reusable page templates.

Panel Editing

Creators should be able to:

- Regenerate one panel.
- Regenerate one character.
- Change expressions.
- Change poses.
- Replace backgrounds.
- Change dialogue.
- Resize panels.
- Merge panels.
- Split panels.
- Insert new panels.
- Remove panels.
- Rearrange page layouts.

No change should require regenerating an entire chapter.

Dialogue System

Dialogue editing includes:

- Speech bubbles.
- Narration boxes.
- Internal monologue.
- Sound effects.
- Font customization.
- Bubble positioning.

- Tail direction.
- Automatic text fitting.

Dialogue remains editable until export.

AI-Assisted Editing

The AI may recommend:

- Better panel pacing.
- Improved camera composition.
- More expressive poses.
- Consistent lighting.
- Dialogue improvements.
- Visual continuity fixes.
- Missing environmental details.

Creators decide whether to accept recommendations.

Asset Reuse

Before generating new content, the Comic Studio shall:

- Search the Asset Library.
- Reuse approved characters.
- Reuse outfits.
- Reuse backgrounds.
- Reuse props.
- Reuse effects.

Only missing assets should be generated.

User Stories

As a creator, I want to edit individual panels without affecting the rest of the page.

As a creator, I want AI to remember my approved character designs.

As a creator, I want to publish in different comic formats without recreating the project.

As a creator, I want complete artistic control while benefiting from AI automation.

Functional Requirements

The Comic Studio shall allow users to:

- Generate pages.
- Edit pages.
- Duplicate pages.
- Archive pages.
- Restore pages.
- Rearrange chapters.
- Export comics.
- Save templates.
- Create reusable layouts.
- Attach creator notes.
- Preview final output.

UI Components

The Comic Studio interface should include:

- Page Canvas
- Panel Editor
- Layer Panel
- Dialogue Editor
- Asset Browser
- Character Browser
- AI Suggestions Panel
- Timeline Panel
- Thumbnail Navigator
- Properties Panel
- Export Panel

Database Entities

Primary entities include:

- Comic
- Chapter
- Page
- Panel
- Dialogue

- PanelAsset
 - LayoutTemplate
 - ComicRevision
 - ExportProfile
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API Considerations

Potential endpoints include:

- Generate Comic
 - Retrieve Page
 - Update Panel
 - Create Chapter
 - Export Comic
 - Save Template
 - Validate Comic
 - Preview Output
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Non-Functional Requirements

The Comic Studio should:

- Handle projects with thousands of pages.
 - Autosave continuously.
 - Load pages efficiently.
 - Render high-resolution artwork.
 - Maintain responsive editing performance.
 - Support future collaborative editing.
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Error Handling

If generation fails:

- Preserve completed panels.
- Resume unfinished tasks.
- Explain the issue.
- Suggest corrective actions.

If consistency conflicts are detected:

- Highlight affected panels.

- Recommend fixes.
 - Allow creator override.
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Future Expansion

Future versions may include:

- Collaborative editing.
 - AI inking.
 - AI coloring.
 - AI lettering.
 - Automatic translation.
 - Interactive comics.
 - Dynamic layouts.
 - Community templates.
 - Marketplace integration.
 - Cross-project asset sharing.
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Acceptance Criteria

The Comic Studio succeeds when creators can efficiently produce, edit, and publish professional-quality comics while maintaining complete creative control and benefiting from intelligent AI assistance.

Key Principle

The Comic Studio is not an image generator. It is a complete comic production environment where AI handles repetitive production work and creators remain the directors of every artistic decision.

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